Using measures of psychophysiological and neural activity to advance understanding of psychological processes in politics

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Abstract

The chapter seeks to inspire the next generation of political psychology research to move beyond self-reports and adopt theories and methods that help us to better characterize the unconscious, affective-cognitive processes that precede political decisions. To this end we provide an overview of work using psychophysiological measures, eye-tracking, and neuroimaging measures. We discuss the opportunities of these measures and how to address the obstacles these measures present. This way we hope to contribute to the promise of a beautiful friendship between political science and neuroscience.

Introduction

Innovation implies successful introduction. Measures of psychophysiological and neural activity have been introduced to political science but the fact they are still rarely used, particularly in mainstream political science, indicates that this introduction has not been successful. Wahlke and Lodge (1972), to our best knowledge, were the first to introduce psychophysiological measures to political science. With only 34 citations (Google Scholar, May 2023) it was not a successful introduction. The path-breaking work of the so-called Dartmouth group about emotional expressions of leaders and facial mimicry (McHugo, Lanzetta, & Bush, 1991; McHugo, Lanzetta, Sullivan, Masters, & Englis, 1985) has had very few follow-ups (for a recent exception, see Homan, Schumacher, & Bakker, 2023). Even the most prominent attempt to link a psychophysiological measure to politics - in this case: skin conductance and ideology (Oxley et al., 2008) - is unlikely to stimulate a wider adoption of psychophysiological measures. This is primarily because failures to replicate the original study (Arceneaux, Bakker, & Schumacher, 2023; Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Fournier, Soroka, & Nir, 2020; Osmundsen, Hendry, Laustsen, Smith, & Petersen, 2022) have stimulated broader discussions about the validity of skin conductance measurements (Osmundsen et al., 2022). Regarding neural activity, over time several people have argued that neuroscience and political science could develop a beautiful friendship (for instance, Cacioppo & Visser, 2003; Jost, Nam, Amodio, & Van Bavel,

2014; Lieberman, Schreiber, & Ochsner, 2003; McDermott, 2009; Zmigrod & Tsakiris, 2021). However, at best, the two fields have barely become acquainted with each other. Instead, the status quo in political science is to rely on self-reported measures of feelings, beliefs, and opinions. A recent review of research on "emotion and politics" in the Annual Review of Political Science (Webster & Albertson, 2022) underlines this point as it contains very few references to work using measures of psychophysiological and neural activity.

We identify three problems with this status quo. First, unconscious (or preconscious) affective processes cannot be measured with self-reports because we are mostly unaware of these processes. This is problematic because several political science theories attribute a role to affect - e.g. motivated reasoning (Lodge & Taber, 2013), affective intelligence (Marcus, Neuman, & MacKuen, 2000) and receive-accept-sample (Zaller et al., 1992) - but do not explicitly measure it, even though we know that affective responses influence our attitudes and behaviors (Bakker, Schumacher, & Rooduijn, 2021; LeDoux & Pine, 2016; Lodge & Taber, 2013; Ryan & Krupnikov, 2021). Second, the complex interplay between affective and cognitive processes at the neural level cannot be characterized by relying on typical (survey) experimental outcome measures. Therefore we know little about the process preceding a decision. Also, we risk assuming that people with the same behavioral or attitudinal outcome have had the same affective-cognitive process leading to the decision. This may not be the case. This becomes problematic if we base interventions on the basis of these behavioral and attitudinal outcomes without understanding the underlying, primarily unconscious affective-cognitive process. Third, on socially sensitive topics, people may be motivated to self-report in socially desirable ways. As such, self-reports may even be misleading. By better capturing the process prior to self-reports we may gain a richer understanding of socially desirable behavior. In sum, political psychology largely concentrates on more conscious reflections at the expense of potentially relevant preconscious or unconscious explanations or mechanisms.

The goal of this chapter is to inspire the next generation of political psychology research to move beyond self-reports and adopt theories and methods that help us to better characterize the affective-cognitive processes of political decision-making. To this end we provide an overview of work using psychophysiological measures (skin

conductance, heart rate variability, and facial electromyography (fEMG)), eye-tracking, and neuroimaging measures (electroencephalography (EEG), functional and structural magnetic resonance imaging). We admit this overview is somewhat all over the place because a lot of work we will review is relevant to politics, but not done by political scientists or published in political science. Also, earlier attempts to develop a specific political science research agenda that integrates the body have failed to become mainstream. Given limits to the length of this chapter and our own knowledge, we recognize that we probably missed important contributions published outside of our scope and ignore other promising measures such as genes (e.g., Aarøe et al., 2021; Dawes & Weinschenk, 2020; Hatemi et al., 2014), hormones (e.g., De Dreu & Kret, 2016; Friesen, Gruszczynski, Smith, & Alford, 2020; Hoyt, Zeiders, Chaku, Toomey, & Nair, 2018; Merolla, Burnett, Pyle, Ahmadi, & Zak, 2013; Zeiders, Nair, Hoyt, Pace, & Cruze, 2020), gut bacteria (e.g., Amato et al., 2021; Greenhough et al., 2020; Johnson, 2020), and taste receptivity (e.g., Friesen, Ksiazkiewicz, & Gothreau, 2021).

In each section we first briefly introduce each measure. Then, we concentrate on two central questions for political psychologists that we identified: first, how can measures of psychophysiological and neural activity help us understand how we process political information with which we disagree as opposed to political information we agree with. We understand political information here broadly as a stimulus that shows a political argument or frame, a politician or party logo, or a combination of both. We use the concepts of incongruent versus congruent stimuli. Congruent stimuli contain information which aligns with the participant, incongruent stimuli contains information which the participant opposes, as typically measured by a pre-experimental survey. The second question is: is there any between-person politically-relevant variation (e.g. ideology, gender, sophistication, etc.) in psychophysiological and neural activity in processing congruent and incongruent political information. We end each section with a reflection on the way forward using this measure. Here we discuss alternative ways in which the measures are used and assess validity concerns. Our focal points in this chapter are skin conductance and fEMG because of our own expertise. We end with an agenda for the next generation of research.

Skin conductance

Skin conductance captures the activity of the autonomic nervous system and particularly the sympathetic nervous system (Dawson, Schell, & Filion, 2017). The autonomic nervous system controls our glands, and skin conductance specifically measures the varying electrical properties of the skin in response to the increase of sweat secretion in the eccrine glands (for an introduction, see Soroka, 2019). With more sweat secretion, the conductance of electricity improves and skin conductance levels rise. This is interpreted as an increase in arousal. In political psychology it is common to measure skin conductance levels by placing electrodes on the fingertips on the non-dominant hand (see panel A of Figure 1 for an example). Through these electrodes, a small current passes onto the skin via the two electrodes. By keeping the current constant it is possible to measure the flow of the current - what we call skin conductance expressed in micro-Siemens (Dawson et al., 2017). Potter and Bolls (2012) also discuss other places on the body where skin conductance can be collected.

Arousal is part of the two-dimensional circumplex model of affect, together with valence (Russell, 1980). Affect can be seen as part of the larger concept of emotion. We follow Keltner and Gross (1999, p. 468) who define emotion as "episodic, relatively short-term, biologically based patterns of perception, experience, physiology, action, and communication that occur in response to specific physical and social challenges and opportunities". Increases in skin conductance, and thus arousal, have been reported in response to negative images (P. J. Lang, Greenwald, Bradley, & Hamm, 1993; Oxley et al., 2008), negative news (Soroka, Fournier, & Nir, 2019; Soroka & McAdams, 2015), threatening news (Boyer, 2021), incivility (Mutz, 2007; Mutz & Reeves, 2005), political rhetoric (Bakker et al., 2021) and negative political ads (Daignault, Soroka, & Giasson, 2013; Wang, Morey, & Srivastava, 2014). At the same time, skin conductance increases in response to positive stimuli such as an exposure to a preferred football team (Potter & Keene, 2012) or politician (Wagner et al., 2015). Arguably the best known research line in political psychology using skin conductance is the literature on ideological asymmetry. In their seminal article Oxley et al. (2008) show that conservatives having strong stronger skin conductance responses to negative stimuli than liberals do. In this research line, physiological arousal is an independent variable and is considered a trait (for similar approaches, see Aarøe, Petersen, & Arceneaux, 2017; Arceneaux, Dunaway,

& Soroka, 2018; Coe, Canelo, Vue, Hibbing, & Nicholson, 2017; Smith, Oxley, Hibbing, Alford, & Hibbing, 2011). Yet, a series of close (direct) replications failed to find indications that there is a latent threat sensitivity dimension and, as a consequence, these projects also could not replicate the association between physiological threat sensitivity and (social) conservatism across different political contexts (Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Fournier et al., 2020; Osmundsen et al., 2022). We think that the failed replications have moved science forward. Yes, there seems no direct, unmediated, effect of physiological responses to threat on ideology. But that should not lead us to disregard the method all together. Skin conductance has been a well-validated measure of the activation of arousal (Settle et al., 2020). Using skin conductance as an indicator of arousal to politically relevant stimuli (political parties, politicians, the news, campaign ads, etc.), seems to be a more fruitful approach than the threat sensitivity approach. Therefore we discuss in the next sections skin conductance as a measurement of arousal instead.

Skin conductance and responses to political information

Returning to our two main questions guiding this chapter: is incongruent political information more arousing than congruent information, and is this incongruence more arousing for some than for others? Wang et al. (2014) report that arguments by opposing candidates are more arousing. In analyzing skin conductance responses to pro-establishment versus anti-establishment rhetoric we also find stronger skin conductance responses to rhetoric that is incongruent with the participant's voting behavior (Schumacher, Rooduijn, & Bakker, 2022). Yet, in a much larger study in which we contrast responses to congruent and incongruent messages about 5 different political issues we do not find this general effect (Bakker et al., 2021)[2]. In another paper, we expose participants to pictures of inparty and outparty politicians (Bakker, Schumacher, & Homan, 2020). We find more skin conductance response to outparty politicians than inparty politicians - but this effect is not statistically significant. In a similar analysis, Petersen, Giessing, and Nielsen (2015) expose participants to logos of political parties. They also find the same negative, but not statistically significant effects we report.

Overall, the direction of the effects in these studies suggests that incongruent political

[2] Note that this result for skin conductance is not in the paper, but we are happy to share it of course

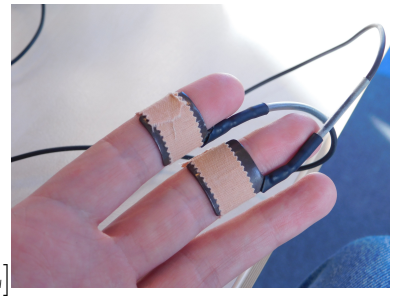
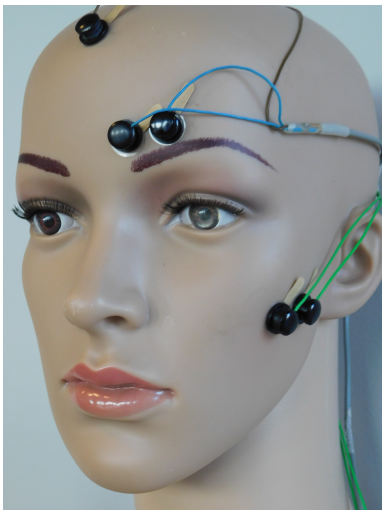
information is more arousing than congruent political information. Yet the evidence is very weak as the tests do not reach statistical significance. From our theoretical point of view one could interpret the difference between incongruent versus congruent information through the lens of the negativity bias (Dubel, Schumacher, Homan, Peterson, & Bakker, 2024; Fournier et al., 2020; Soroka et al., 2019; Soroka & McAdams, 2015). Across 17 countries on 6 different continents there is uniform evidence that negative news is more arousing than positive news (Soroka et al., 2019). Recently, we succesfully replicated this finding in the Netherlands and showed that negative news also causes more corrugator activity (as a measure of negative valence) (Dubel et al., 2024). If we interpret incongruency as negative, then the negativity bias should extend to how we process ingroup and outgroup messages and candidates. Yet, whether incongruent arguments are indeed conceptually similar to negative news can be discussed. For one, incongruent arguments can still have a positive or a negative emotional framing. As such, both the position and the framing of arguments produce affective responses (Daignault et al., 2013), possibly obfuscating a clear relationship between arousal and incongruence.

Also, Soroka et al. (2019) suggest that there is still a lot of between-subject variation in the strength of the negativity bias. This brings us to the second question guiding this chapter: is there between-subject variation in how we respond to congruent and incongruent political information? In processing congruent and incongruent arguments about different issues, we find that people with more extreme political attitudes experience more arousal, regardless of the ideological slant of the argument. In the same study, we find a positive but, not statistically significant, association between political knowledge and arousal (Bakker et al., 2021). We agree with Soroka et al. (2019) that systematic research efforts are needed to explain between-subject variation in physiological arousal.

To summarize, SCL has been quite extensively studied in political psychology compared to other measures we discuss in this chapter. SCL equipment is not very expensive and SCL is relatively easy to collect (placing electrodes on fingers is relatively straightforward). Using SCL as an indicator of a physiological disposition (e.g., threat sensitivity) is, most likely, not the right way to conceptualize SCL in political psychology: there is not physiological trait *and* these traits have no predictive validity.

Yet, that does not meant that SCL should be abandoned. Instead, we think there is still a lot of potential of exploring SCL responses as indicators of arousal in response to political communication, for instance. At the same time, this requires careful designs that clearly isolate the cause of an effect. This is not easy but that should not let us shy away from using the measure. Finally, as with all measures discussed in this chapter, SCL comes with a lot of researcher degrees of freedom, so we welcome more preregistered studies. Robustness checks across different specifications could also help solve some emerging discussions about how SCL should *best* be modelled.

Figure 1. Physiological measures: SCL, EMG

[A: SCL]  [B: EMG] 

Note: Picture A shows skin conductance measurement using the finger placement. The hand belongs to the first author of this chapter. Picture B shows the facial electromyography setup, with the electrodes on the cheek measuring zygomaticus activity and the electrodes above the brow measuring corrugator activity. The top black electrode is a ground electrode. Picture B shows Sensi, our demonstration doll. Before joining our lab she had a distinguished career in the fashion industry,

Facial electromyography

Political psychologists (1) sometimes acknowledge they "lack direct measures of somatic response" while they do theorize about affect (Lodge & Taber, 2013, p. 208), or (2) only study one dimension of core affect, specifically, arousal (Carlson, McClean, & Settle, 2020; Mutz, 2007; Soroka & McAdams, 2015), or (c) study the cognitive outcomes of affect (Keltner & Gross, 1999), such as discrete emotions like anger and anxiety (Aarøe, 2011; Huddy, Mason, & Aarøe, 2015; Klar, 2013). As we discussed in the previous section, arousal has been studied quite extensively by political

psychologists. We think fEMG offers largely unexplored potential as it captures valence by registering rapid, partially automatic affective responses in the face while participants view stimulus material.

fEMG measures tiny, partially invisible, rapid, bioelectric signals at the surface of the skin that are generated due to muscle contractions in a specific region (Blascovich, Vanman, Mendes, & Dickerson, 2014; Bucy & Bradley, 2011; Potter & Bolls, 2012; Tassinary, Cacioppo, & Vanman, 2007; van Boxtel, 2010). fEMG is primarily used as a measure of emotional valence (Larsen, Berntson, Poehlmann, Ito, & Cacioppo, 2008; Tassinary et al., 2007). The so-called *corrugator supercilii* muscle region – the muscle above the eye-brow that draws the brow down and pulls the brows together – registers negative affect. Indeed, corrugator activity has been recorded in response to negative images (P. J. Lang et al., 1993), negative words, negative affective cues (Hietanen, Surakka, & Linnankoski, 1998), negative news (Boyer, 2021), The so-called *zygomaticus major* muscle region, which pulls the corners of the mouth up and back into a smile (Larsen, Norris, & Cacioppo, 2003), registers a positive affect. Indeed, zygomaticus activity increases in response to positively valenced images (Van Oyen Witvliet & Vrana, 1995) and videos (Cacioppo, Petty, Losch, & Kim, 1986). fEMG is collected by placing electrodes on the specific muscle area - see Figure 1, panel B for an example. The corrugator is measured by placing two electrodes above the eye-brow, while the zygomaticus is measured by placing two electrodes on the cheek. The fifth electrode on the image is a ground electrode.[3]

fEMG is by no means limited to studying the corrugator and zygomaticus. Other muscles that can be captured using fEMG have been associated with specific discrete emotions: the *levator labii* is associated with disgust (Chapman, Kim, Susskind, & Anderson, 2009), while the *frontalis* is associated with fear. There are concerns whether the *levator labii* and *frontalis* validly and reliably capture these distinct emotions (Tassinary et al., 2007), recent advances in, particularly, data science hold the promise of detecting discrete emotions using fEMG (Sharma, Castellini, van den Broek, Albu-Schaeffer, & Schwenker, 2019).

Using a Google Scholar search, we found only 18 studies in peer-reviewed journals

[3] Schumacher, Homan, et al. (2023) provides overview of how to process and analyze fEMG work set in political science.

that use fEMG in the domain of politics (Schumacher, Homan, et al., 2023). Table 1 provides an overview and discuss some of these studies in this section. A few general patterns stand out in this table: fEMG has been mostly used in studies published in other fields than political science. Second, the corrugator is the most widely studied muscle and the sample sizes vary tremendously: from 21 (Marcus, Wood, & Theiss-Morse, 1998) to over 300 (Peterson, Jacobs, Hibbing, & Smith, 2018).

fEMG and responses to political information

Returning to the key questions of this chapter, we found that incongruent arguments produce more corrugator activity (negative affect) than congruent arguments (Bakker et al., 2021). We found no difference in zygomaticus activity between incongruent and congruent arguments. Self-reported discrete emotions were only very weakly correlated with the physiological responses to incongruent and congruent messages. In our much smaller study contrasting responses to antiestablishment versus proestablishment rhetoric we find very weak and inconsistent evidence for this incongruency effect on corrugator activity (Schumacher et al., 2022). Also Wang et al. (2014) do not report a general negative affective response to incongruent arguments, but they use actual campaign ads from candidates in their experiment, so that one cannot separate responses to arguments from responses to candidates. Finally, Goudarzi, Pliskin, Jost, and Knowles (2020) found that those higher on system justification experienced *less* corrugator activity when exposed to images of homeless people, compared to those lower on systematic justification. This led Goudarzi et al. (2020, p.2) to conclude that "exposure to extreme manifestations of inequality (e.g., people experiencing homelessness) elicits less negative affect in people who regard the prevailing economic system as more (vs. less) fair and legitimate". We can interpret this in the context of congruence-incongruence: for people scoring low on system justification, inequality is more incongruent and thus produces more corrugator activity.

There is more work measuring fEMG responses to political leaders (Bakker, Schumacher, & Homan, 2020; Bourgeois & Hess, 2008; Fino, Menegatti, Avenanti, & Rubini, 2019; Homan, Schumacher, & Bakker, 2023; McHugo et al., 1991, 1985). The so-called "Dartmouth group" studied the affective responses to emotional displays from politicians. McHugo et al. (1985), for instance, measured physiological responses

(corrugator and zygomaticus) and self-reported emotions to politicians' emotional displays. They find that participants showed increased corrugator activity and self-reported negative emotions in response to the anger and fear displays, while they showed increased zygomaticus activation and reported more positive emotions in response to the happiness displays. Yet, across studies the extent to which existing attitudes condition the physiological responses to politicians differs. In one study McHugo et al. (1985) find that people's attitudes towards the politicians that displayed the emotion (e.g., Ronald Reagan) were not affected by participants' prior attitudes towards the politician, while McHugo et al. (1991) find that prior attitudes condition the affective responses. Specifically, in an experiment where emotional displays of President Reagan and Senator Gary Hart (Reagan's opponent) were used as stimuli, they find that people with favorable attitudes towards the politicians showed more zygomaticus activity in response to the politicians' happiness displays. In yet another study, Bourgeois and Hess (2008) find that prior attitudes towards politicians do not condition the zygomaticus responses, but instead it conditions how they respond to angry displays: supporters of the politician showed more corrugator activity compared to opponents.

The studies from the Dartmouth group have shortcomings: their samples sizes were small (N=40 to N=100) providing limited statistical power to test between-subjects conditions (e.g., prior attitude, display modality). Including only one or two politicians limits generalizability and could bias the results. Finally, the studies discussed here did not have complete experimental control over the expressions of politicians since the displays in these studies consist of video clips selected from televised debates and speeches. As such, the intensity, onset, and offset of the emotional expressions in these clips might vary (Sullivan, 1996), making the within-person responses to a supporting and an opposing politician less comparable.

Addressing these shortcomings, and building on this work Homan, Schumacher, and Bakker (2023) compared participant fEMG responses to images of normal citizens, the inparty leader and the outparty leader, displaying either an anger, happy or a neutral expression. The study was well-powered to detect small effects, relied upon multiple politicians, and had experimental control over the intensity of the emotional displays as these were manipulated using state-of-the-art computer software and validated by

independent coders. Outparty leaders - compared to inparty leaders - produce more corrugator activity overall, and also more zygomaticus activity in the anger condition (i.e. a reactive response). Surprisingly, participants relaxed their *corrugator* muscle and had increased *zygomaticus* activity in response to inparty politicians' emotional displays, no matter what emotion the inparty politician displayed. In all, whether the politician was inparty or outparty was more important than the emotional expression. These findings are in line with the emotional mimicry in social context model: happy displays by inparty politicians are mimicked, but displays with outparty politicians are generally met with negativity.

To conclude, we think fEMG has a lot of potential in political psychology but has been underused. The equipment to collect fEMG is not very expensive and it is not that hard to learn. Yet, compared to heart rate and SCL, fEMG is harder to collect: the skin needs to be cleaned and multiple electrodes need to be placed at specific places on the face. A lot can go wrong here, leading to unreliable or invalid measures of fEMG. Moreover, analysing fEMG is not easy: outliers are present and cross-talk pollutes the measurement. Careful and transparent preprocessing procedures are necessary working with fEMG. All in all, it seems quite daunting to collect fEMG. Yet, we think that the studies we discussed here provide evidence that fEMG is meaningful for political psychology. Moreover, theories of affect all point out that arousal is just one dimension of affect. If we are interested in affect, then we should theorize and measure more than just arousal. We therefore encourage others, like us, to start using fEMG in their research.

Heart Rate (Variability)

Heart rate variability is in some cases used to measure arousal, and in other cases, it is used to measure (cognitive) attention (A. Lang, Newhagen, & Reeves, 1996; Soroka et al., 2019; Soroka & McAdams, 2015). It is argued that any acceleration in heart rate that comes from arousal will be overwhelmed by the deceleration that comes with attention (Potter & Bolls, 2012). Perhaps this combination of initial arousal that causes increase in heart rate followed by a decrease in heart rate because of cognitive attention causes the strongest increase in heart rate variability. Heart rate is measured, relatively easily, by placing electrodes on the skin around the heart or, even easier, with a

Table 1
Overview of studies on politics using fEMG measures

Authors	Journal	Main title	fEMG measures	N
McHugo et al. (1985)	JPSP	Emotional reactions to a political leader's expressive displays	Corrugator, zygomaticus	40
McHugo et al. (1991)	JNB	The effect of attitudes on emotional reactions to expressive displays of political leaders	Corrugator, zygomaticus	100
Marcus et al. (1998)	PLS	Linking Neuroscience to Political Intolerance and Political Judgment	Corrugator	21
Ensari et al. (2004)	GPIR	Negative Affect and Political Sensitivity in Crossed Categorization	Corrugator	45
Bucy and Bradley (2004)	SSI	Presidential Expressions and Viewer Emotion	Corrugator, zygomaticus	41
Oxley et al. (2008)	Science	Political attitudes vary with physiological traits	Orbicularis oculi	48
Bourgeois and Hess (2008)	BioPsy	The impact of social context on mimicry	Corrugator, zygomaticus, levator labii, orbicularis oculi	104
Wang et al. (2014)	ComRes	Motivated Selective Attention During Political Ad Processing	Corrugator, zygomaticus	120
Bradley, Angelini, and Lee (2007)	Jadv	Psychophysiological and Memory Effects of Negative Political Ads	Orbicularis oculi	51
Peterson et al. (2018)	PLS	Emotional expressivity as a predictor of ideology	Corrugator	342
Fino et al. (2019)	SciRep	Unfolding political attitudes through the face	Corrugator, zygomaticus	53
Bakker, Schumacher, and Homan (2020)	PLS	Yikes! Are we disgusted by politicians?	Levator labii, corrugator	108
Bakker, Schumacher, Gothreau, and Arceneaux (2020, NL sample)	NHB	Conservatives and liberals have similar physiological responses to threats	Corrugator	81
Bakker, Schumacher, Gothreau, and Arceneaux (2020, US sample)	NHB	Conservatives and liberals have similar physiological responses to threats	Levator labii, corrugator	202
Goudarzi et al. (2020)	NatCom	Economic system justification predicts muted emotional responses to inequality	Corrugator, levator labii	155
Bakker et al. (2021)	APSR	Hot politics? Affective responses to political rhetoric	Corrugator, zygomaticus	397
Boyer (2021)	IntJPP	How the News Exacerbates Motivated Reasoning	Corrugator	191
Schumacher et al. (2022)	Pol Psy	Hot populism? Affective responses to antiestablishment rhetoric	Corrugator, zygomaticus	343
Homan, Schumacher, and Bakker (2023)	Emotion	Do Emotional Displays of Politicians Evoke Mimicry and Emotional Contagion?	Corrugator, zygomaticus	107

standard heart rate band as used by athletes.

Heart rate and responses to political information

In communication science research, heart rate has been used to study attention of audience members. Most of this work builds upon the limited capacity theory (A. Lang, 2000) which holds that attention is under control of both the receiver (the participant) and the message characteristics. Receivers can, actively, allocate attention to a message. Yet, characteristics of the message – for instance whether it is negative or features an (outparty) politician, etc – can also evoke attention as a more automated reflex. Scholars in communication research have used heart rate as a way to capture this more reflective attention: whereby the heart rate decreases as the attention to message increases (Potter & Bolls, 2012). A. Lang et al. (1996) found that heart rate decreases in response to negative news vs. positive news (for a similar type of study, see Grabe, Zhou, Lang, & Bolls, 2000).

Heart rate has made its way in the political psychology literature. Most prominently, in a seminal study, Soroka and McAdams (2015) find that heart rate is lower in negative vs. positive news. The authors interpret the stronger attention to negative news as an indication of a negativity bias. There seems to be heterogeneity in this negativity bias: Soroka, Gidengil, Fournier, and Nir (2016) find that women have a

stronger decrease in heart rate in response to negative vs. positive news compared to men. This led Soroka et al. (2016, p. 361-362) to conclude that "that women pay more attention to negative news stories than men do". The studies discussed before in this paragraph were conducted in 1 country (US or Canada). In their monumental study, Soroka et al. (2019, p. 18889) rely upon heart rate variability (the root mean square of the successive difference [RMSSD] to be specific) which captures "a combination of activation (increasing heart rate) and attentiveness(decreasing heart rate)". They find that negative news causes an increase in heart rate variability compared to negative news. The authors interpret this as evidence that negative news is capturing more attention compared to positive news. In another study, Dunaway and Soroka (2021) asked whether information processing differs when people consume information on a smartphone or on a larger computer screen. Dunaway and Soroka (2021, p. 69) conclude that the "results suggest lower levels of cognitive access" as captured with heart rate variability "to video news content on a mobile-sized screen" compared to the normal screen. Dunaway and Soroka (2021, p. 69) conclude that their study has "potentially important consequences for public attention to current affairs in an increasingly mobile media environment."

In line with the discussion in our paragraph on skin conductance, we may hypothesize that heart rate variability should also increase in response to incongruent political stimuli. Yet, in a more direct test we found no evidence that heart rate variability differs depending on whether the message shown in the experiment was pro or counter attitudinal (Bakker et al., 2021). Such null-findings are equally important to document and we return to this point in the discussion. Another study that speaks to this question was conducted by Carlson et al. (2020). In their study Carlson et al. (2020, p. 176) informed people that they were about to have a political discussion using the prompt "in the next part of the study, you will discuss a controversial political topic with another study participant. Before you meet your discussion partner, we will give you a little information about them based on their online survey responses." The increase in heart rate while reading this prompt, compared to a pre-treatment baseline, was correlated with the homogeneity of the participants' actual social network. Carlson et al. (2020, p. 165) find that "people who experienced an increase in heart rate when they were informed they were more likely to be in homogeneous discussion networks".

This association suggests according to Carlson et al. (2020) that "those who are more physiologically reactive to anticipating a political discussion are more likely to discuss politics with copartisans" and that the "physiological experience of interacting with disagreeable others may be a factor influencing the choices people make in structuring their discussion networks." Returning to our question, Carlson et al. (2020) suggest that physiological responses correlate with the extent to which people expose themselves to incongruent political information.

Alternative connections between heart rate and political psychology

Political psychology has almost solely studied heart rate as a measure of attention. We want to outline two possible other avenues for scholars that rely upon heart rate - broadly conceived. First, recent work by Tsakiris, Vehar, and Tucciarelli (2021) propose an entirely different way to use heart rate. Arterial baroreceptors signal cardiovascular arousal to the brain, and in this way strengthen fear and anxiety. These baroreceptors, however, only fire during the systolic phase of the cardiac cycle and not during the diastolic phase. By time-locking treatments to one of the two phases, we can more closely investigate the role of fear and anxiety in political decision-making (Tsakiris et al., 2021).

Second, the BioPsychoSocial model of Challenge/Threat (BPS-CT) (Blascovich, 2013) could be of interest for political psychologists. The BPS-CT assumes that motivated performance situations are evaluated by people based upon the demands (e.g., uncertainty, required effort) and the available resources to deal with these demands (e.g., skills). As Domen, Scheepers, Derks, and van Veelen (2022, p. 713) explain, "the ratio of perceived demands and resources either results in a state of challenge (i.e., resources outweigh demands) or a state of threat (i.e., demands outweigh resources)." Both challenge and threat can be evaluated as negative or positive forms of stress. And this stress can be captured by a more or less efficient cardiovascular response pattern. Scheepers and Derks (2016) have made the case that high social power causes a different cardiovascular response they call "challenge", while the activation of low social power elicits a cardiovascular response they call "threat". According to this line of research, threat and challenge can be distinguished using Cardiac Output (CO) - the amount of blood the heart pumps around in 1 minute - and

the Total Peripheral Resistance (TPR) - the resistance of blood vessels to blood flow (Behnke & Kaczmarek, 2018). Threat is indicated by low CO and high TPR, while challenge is indicated by high CO and low TPR (Behnke & Kaczmarek, 2018).

In a recent study, Domen et al. (2022) conclude that "men with relatively high gender identification showed the strongest tendency towards threat when presented with a woman who challenged the gender hierarchy, and the strongest tendency towards challenge when presented with a woman who legitimized the gender hierarchy" Domen et al. (2022, p. 716). In an earlier study Scheepers, Röell, and Ellemers (2015) find that participants in a stable high-power position showed a cardiovascular response pattern that was indicative of (positive) challenge. Yet, when participants in a high-power position were in an unstable position, they showed a cardiovascular response pattern that was indicative of threat. Cardiovascular responses patterns associated with threat also correlate with poor mental health (Blascovich et al., 2014). While speculative, this could perhaps be a path-way into the link between politics and mental health (Smith, 2022; Tsakiris et al., 2021).

As heart rate, and other measures of cardiac activity, are not very costly, nor very difficult to collect (analysing is, in line with all measures described here, more complicated), we hope to welcome more research that theorize, test and report the results for heart rate in the field of political psychology. Again, like SCL and fEMG, this does require designs that can carefully disentangle cause from effect.

Eyetracking

Eyetracking can be used to measure visual attention to a stimulus or to different objects within a stimulus (Kohout, Kruikemeier, & Bakker, 2023; Ohme, Searles, & de Vreese, 2022). Attention does not equate awareness (Lamme, 2003). People are often unaware of the location of their fixations (Hannula et al., 2010), and also their eye movements can be guided memories people are unaware of (Tanenhaus, Magnuson, Dahan, & Chambers, 2000). As such, it can be a highly relevant tool, but as of yet it is more common in communication research (Dunaway, Searles, Sui, & Paul, 2018; Geise, Heck, & Panke, 2021; Geise, Panke, & Heck, 2021; Keib et al., 2022; Ohme, Maslowska,

& Mothes, 2021) than in political psychology (or political science).[4]

Eye-tracking data is relatively easy to collect by connecting an eye-tracker to a computer screen (Kohout et al., 2023). By now eye-trackers can be so small that they can be placed on a laptop. Moreover, eye-trackers are also available as glasses to track the consumption of political information when reading news papers (Kruikemeier, Lecheler, & Boyer, 2018) or mounted on a stand to track how people browse their mobile equipment (Ohme et al., 2021). In eye-tracking studies, dwell time is commonly used as indicator for visual attention (Orquin & Holmqvist, 2018). Areas of interest (AOI) are selected regions of a displayed stimulus.

The few applications in political psychology use eye-tracking data differently. First, some work describes ideological asymmetries in attention to positive and negative stimuli(Mills, Smith, Hibbing, & Dodd, 2014; Oosterhoff, Shook, & Ford, 2018). Second, in combination with an innovative experimental design, eye-tracking is used to track gender and partisan biases (Coronel & Federmeier, 2016; Coronel, Moore, & deBuys, 2021; Coronel, Poulsen, & Sweitzer, 2020) in political information processing. Third, eye-tracking is used to study political information processing of news (Dunaway et al., 2018; Kruikemeier et al., 2018) as well as emotionally valenced user comments in response to news (Kohout et al., 2023). Fourth, eye-tracking has been used to validate experimental designs (Jenke, Bansak, Hainmueller, & Hangartner, 2021).

Eye-tracking and responses to political information

Do we pay more attention to politically congruent versus incongruent stimuli? To our knowledge there is not a direct answer to this. There are a few suggestions. Jenke et al. (2021) used eye-tracking in a conjoint leader choice task: a typical setup for such an experiment is that participants are shown two profiles of politicians and are asked to make a choice. These profiles consist of various attributes (e.g. ideology of the candidate or their background). Eye-tracking allows the researcher to analyze the link between visual attention to the attributes in the profile and the outcome variable typically in the form of candidate choice or candidate liking. Jenke et al. (2021) report that participants pay more visual attention to the candidates they like, and the features

[4] A. J. King, Bol, Cummins, and John (2019) provide recommendations and best practices from this field.

they prefer. This would suggest we pay more visual attention to politically congruent stimuli. Also the eye-tracking studies on bias suggest participants pay more attention to congruent stimuli than incongruent stimuli (Coronel, Bullock, et al., 2021; Coronel & Federmeier, 2016). Yet, a more direct test is warranted.

For one, a cornerstone of the motivated reasoning literature is the finding that we pay more time reading and counter arguing against incongruent arguments (Lodge & Taber, 2013). We may pay more attention to incongruent world views because they are potentially more threatening (Soroka & McAdams, 2015). More direct evidence comes from the eye-tracking study by Kohout et al. (2023) who found that people have more visual attention to negative vs. positive messages.

It is likely that visual attention to congruent or incongruent stimuli differ depending on participant characteristics. For example, Coronel and Federmeier (2016) reports that political knowledge is an important moderating factor of visual attention. Yet, we are not aware of studies testing this.

We think the eye-tracking, like fEMG, is under-used in the field of political psychology. In general eye-tracking equipment is not very expensive and the data collection and analysis is relatively straightforward. For sure, we would welcome more research that focuses on eye-tracking in the study of political information processing. Attention is thereby a key concept. Yet, we would like to highlight that an added benefit of collecting eye-tracking data is that most devices simultaneously collect data on pupil size and blinks, which can be used to measure emotional arousal and mimicry (Kret, 2018).

Electroencephalography (EEG)

EEG is a non-invasive neuroimaging technique used to measure and record the electrical activity of the brain. EEG recordings are obtained by placing multiple electrodes on the scalp, as is shown in Figure 2. These electrodes detect and amplify electrical signals produced by the brain's neurons. This sounds simple, but EEG is, compared to some of the other measures we described in this chapter, relatively difficult to collect. A cap is placed on the participants' head and 32-128 electrodes are placed on designated areas indicated by holes in the hat (see Figure 2). Image C, for example,

shows a participant that is ready to start the EEG study.[5] The cap ensures that the researcher places each electrode on the right place. Depending on the research question and the EEG equipment, 32, 64, or even 128 electrodes are used. As one can imagine, this is a delicate process which takes time: in our lab the reusable electrodes need to be filled with a gel, placed on the cap and the research assistant needs to make sure each electrode is correctly capturing the signal. In our Hot Politics Lab it took trained lab-assistants approximately 30 minutes to correctly mount the EEG equipment on the research participant. Compared to heart rate, eye-tracking, and even, fEMG this is a lot of time which asks patience of participants and an extra investment for the research team as both the research assistant and participant need to be financially compensated for their time.

Once the electrodes are in place, the researcher can collect so-called event-related potentials (ERPs). ERPs are patterns of neuronal activity in response to a time-set stimulus introduced by the researchers. This activity typically takes the form of a wave with positive and negative deflections. Typical ERPs are called P300 or N400, referring to a positive deflection (P) or negative (N) deflection, and the timing of this peak in milliseconds (e.g. 300 or 400). P300 is indicative of attention, categorizing and salience, while N400 is indicative of processing semantic and affective incongruity. There are more ERPs that index different processes, sometimes with a different labelling indicating the nth deflection (e.g. N1 or P3) or are more design-based (Error-related Negativity).

Aside from ERPs researchers can also measure neural oscillations or rhythms using EEG. These oscillations also reflect neuronal activity in wave-like form. Yet, these waves operate at different frequencies, and this can be used to distinguish different types of brain waves (e.g. alpha, gamma, beta, mu) that are associated with different psychological states.

EEG is rarely used in political science, yet neuroscientists occasionally use political stimuli in an EEG study (e.g., Galli et al., 2021; Morey, 2017; Vecchiato et al., 2014; Zhang, Sherwin, Dmochowski, Sajda, & Kender, 2014).[6] For example, in a study on the

[5] In the case of this image, it is Maaike Homan - one of our Hot Politics Lab members - who is about to pilot her own EEG-experiment.

[6] EEG is more common in communication research and advertising research. It is beyond the scope of this chapter to review this literature. Yet, we refer to recent reviews in communication (Morey, 2018)

effects of negative (vs. positive) economic news, Petropoulos Petalas, Bos, Hendriks Vettehen, and van Schie (2020, p. 977) find that "economic forecasts ... influence individuals' risk taking and their ERPs to decision outcomes, whereby unexpected outcomes were processed more attentively than outcomes that were in line with previous economic forecasts." According to Petropoulos Petalas et al. (2020, p. 977), "these findings confirm the self-fulfilling prophecy effect in economic forecasting and corroborate existing models that have suggested a causal relationship between economic news and economic decision-making at a macro level. Another study shows that early EEG responses to pre-electoral survey items reflect political attitudes an predict voting behavior (Galli et al., 2021). In another study, Morey (2017) studies the extent to which partisanship moderates the extent to which different ads (positive, negative, or neutral) are remembered in an EEG study. The behavioural data shows that Republicans tend to remember more from positive relative to negative ads, while Democrats remember more from negative ads compared to positive ads (Morey, 2017). Turning to the EEG data, Morey (2017) concludes that there is a direct - unmoderated - effect of the gamma frequency band on memory. What does this gamma frequency band say? According to Knyazev (2007), EEG frequency bands provide information about specific cortical information processes, such as behaviour, emotion, memory formation, attention or perception. The gamma band frequency range has been an indicator of semantic processing. Morey (2017) explains that deeper processing (as indicated by the gamma frequency) is positively associated with better memory. In addition, Vecchiato et al. (2014) show that neuroelectrical correlates of trustworthiness and dominance judgments relate to the observation of political candidates. These three studies illustrate that EEG provides a foundation for our understanding of how people process political candidates.

EEG and responses to political information

The behavioral results of the motivated reasoning framework are underpinned by a relatively little-known study in which ERPs were recorded while participants processes congruent and incongruent information (i.e. arguments, candidate names) (Morris, Squires, Taber, & Lodge, 2003). The study showed rapid affective responses to the

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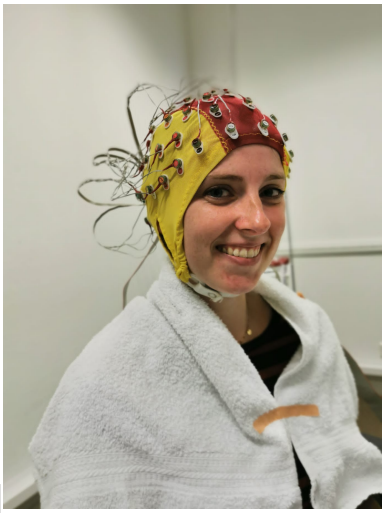
and advertising research (Alsharif et al., 2021).

stimuli prior to processes of cognitive assessment. Specifically, the study identified stronger N400 activity in incongruent information compared to congruent information. N400 is identified with processing semantic and affective incongruity. The study failed to identify differences in P300, which is associated with event probability or salience. In a similar study assesssing ERPs to value-consistent and value-inconsistent words and phrases (Van Berkum, Holleman, Nieuwland, Otten, & Murre, 2009), the N400 effect is replicated. Their interpretation however is different: they consider N400 a consequence of difficulty in processing the meaning. Value-inconsistent words - conceptually similar to what we have called incongruent political information - may be more difficult to process because we expect to hear something else, based on the values we associate with. Yet, the study also identified what is called a late positive potential (LPP) which reflects "affect-induced intensified processing of motivationally important stimuli" (Van Berkum et al., 2009, p. 1093).

Moving from analyzing ERPs to neural oscillations, Homan, Hamdan, Hendriks, and Petropoulos Petalas (2023) analyze the so-called mu and alpha rhythms in participants in response to different emotional displays of inparty and outparty politicians. The mu rhythm is a neural signal in the sensorimotor cortex that is related to the processing and understanding of other people's actions, intentions, and emotions. Mu rhythm is suppressed (decreasing) when sensorimotor cortical activity increases causing the respective neurons to fire out of synchrony. This may happen when people imagine or undertake a motor action. Homan, Hamdan, et al. (2023) find that displaying outparty politicians, compared to inparty and non-politicians causes stronger mu suppression in participants, particularly if the outparty politicians shows anger. They also report more alpha suppression to outparty politicians, particularly when they show happiness (Homan, Hamdan, et al., 2023). This is indicative of increased attention to a stimulus. These findings could be explained by the motivated reasoning framework where incongruent political information receives more attention compared to congruent information. Also, the findings may suggest we prepare more for action if we hear incongruent political information. When it comes to explaining variation in response to congruent and incongruent political information, Homan, Hamdan, et al. (2023) do not find systematic associations between mu rhythm, and political engagement nor self-reported empathy.

The studies discussed here suggest that EEG – as a way to measure brain activity – can help increase the understanding how people process political information. We recognize that EEG is not "easy" to employ for political psychologists. First, it requires specific laboratory equipment which is not "cheap". Second, to get sufficiently powered EEG studies researchers need to use within-subjects with a relatively large number of observations within each individual (Baker et al., 2021). This type of designs are, arguably, less common for political psychologists but are making their way into the field (Homan, Hamdan, et al., 2023; Morey, 2017). Third, after data collection, preprocessing and analysis stage require the researcher to make a lot of decisions (Rodrigues, Weiß, Hewig, & Allen, 2021). It is safe to say that bringing EEG into the toolbox of political psychologists comes with a steep learning curve. Yet, we believe that the benefits of EEG for increasing our understanding about political information processing outweigh the costs.

Figure 2. Neuroimaging measures: EEG, fMRI

[C: EEG]  [D: fMRI] 

functional Magnetic Resonance Imaging (fMRI)

fMRI is a method that utilizes the magnetic properties of hemoglobin to measure the difference between oxygenated and deoxygenated blood (BOLD), providing an indirect assessment of neural activity with high spatiotemporal resolution. In political psychology research, before conducting an fMRI study, investigators usually identify specific brain regions (Regions Of Interest [ROIs]) or networks of ROIs that are known to correlate with a particular psychological concept of interest, such as aggression,

valuation, or attention (although there are exceptions to this practice).[7] fMRI has several advantages over the previously described psychophysiological measures. In particular, fMRI provides the researcher a multivariate recording of a participants' neural activity. Recent work has shown that concepts of interest to political psychologists like discrete emotions (Kragel, Reddan, LaBar, & Wager, 2019) as well as moral intuitions can be "decoded" from fMRI data by combining "patterns" of neural activity within ROIs (or the whole brain). This approach is called a Multivariate Pattern Analysis (MVPA, Norman, Polyn, Detre, & Haxby, 2006). The MVPA allows researchers to move beyond the broader affective dimensions of valence and arousal which psychophysiological detect.

fMRI is collected in a MRI scanner. Figure 2, panel D, is an example of an MRI scanner. This is the 3 Tesla MRI scanner (Phillips Achieva DS, 32 channel head coil) that we have to our availability as a shared research facility at the University of Amsterdam. As can be seen on the image, a participant needs to lay down, and specially trained research personnel fixates the head so that movement is minimal. Afterwards, the participants slides into the cylinder (the MRI scanner), the research personnel takes place in the designed control room, and the scanning can start.

The field of social neuroscience has made a lot of progress using fMRI to understand prejudice - a key outcome of interest in political psychology. It is beyond the scope of this chapter to review this evidence. Instead, we refer to the recent review by Amodio and Cikara (2021). Here, we make the arbitrary decision, to turn to other contributions of fMRI to political psychology. For one, several papers use structural brain scans to investigate correlations between ideology and sizes of particular regions. Kanai, Feilden, Firth, and Rees (2011), for example, reported that liberals have larger gray matter volume in the anterior cingulate cortex (ACC) than conservatives do, but less volume in the right amygdala (rAMG) than conservatives. Follow-up work identified more ROIs related to ideology, and also analyzed ideological differences in brain activity during task performance (Kanai et al., 2011; Knutson, Wood, Spampinato, & Grafman, 2006; Krosch, Jost, & Van Bavel, 2021; Nam, 2020; Nam, Jost, Kaggen, Campbell-Meiklejohn, & Van Bavel, 2017; Schreiber et al., 2013; Yang, Wilson, Lu, & Cranmer, 2022). Yet,

[7] Levy and colleagues (Levy, Goldstein, Influs, Masalha, & Feldman, 2022; Levy et al., 2016) used Magnetoencephalography (MEG) to study intergroup conflict. While MEG is beyond the scope of this book chapter, MEG is definitely worth the attention of political psychologists.

many of these studies suffer from low-N and low variation in ideology. For example, the Kanai et al. (2011) study has not a single participant at point 5 of their 1-5 scale measuring conservatism. A recent working paper analyzes the largest (N=918) and most diverse sample size to date and replicates the finding that amygdala size is weakly, positively correlated with conservatism but finds no evidence for the other ROIs identified in earlier studies (Schumacher, Petropoulus Petalas, & Scholte, 2023).

fMRI and responses to political information

What can fMRI tell us about how we process politically congruent or incongruent stimuli? Westen, Blagov, Harenski, Kilts, and Hamann (2006) reported that in processing political arguments brain regions associated with (unconscious) emotional processing were activated, whereas regions associated with reasoning processing and emotional regulation were not. Follow-up research found similarly stronger activation of the anterior cingulate cortex and insular cortex in comparing incongruent versus congruent arguments (Haas, Baker, & Gonzalez, 2017, 2021). Yet, liberals have the strongest activation if in-group politicians present incongruent arguments, whereas conservatives have the strongest activation if out-group politicians present incongruent arguments (Haas et al., 2017). In addition, also the certainty with which a politician in the experiment conveys their position, stimulates stronger ACC and left insula activation (Haas et al., 2021).

Kaplan, Gimbel, and Harris (2016) contrasted how people processed political beliefs they disagreed with and evidence supporting these views with non-political beliefs. They report that in response to political vs non-political beliefs there is relatively more processing in the Default Mode Network (DMN) areas including the precuneus, the posterior cingulate cortex, the medial prefrontal cortex, the inferior parietal lobe, and the anterior temporal lobe and less activity in the dorsolateral prefrontal cortices and the orbitofrontal cortices. The latter finding seems to relate to the lower rate of belief change in political versus non-political beliefs in the experiment (Kaplan et al., 2016). Also they report - like the studies we discussed in the previous paragraph - the involvement of the insula in resistance to belief change. The activation of the DMN areas is according to authors indicative of the processing of deeply-held beliefs (Kaplan et al., 2016).

Asking people to evaluate different moral violations, recent work by Hopp et al. (2023), shows that political ideology moderates the responses to different moral foundations. In particular, Hopp et al. (2023) find that the neural responses of liberals and conservatives differ when they are asked to evaluate individualizing versus binding moral foundations. To be specific, ideology moderates the neural responses to individualizing vs. binding foundations in the lingual gyrus (LG), V1, anterior prefrontal cortex (aPFC), and superior temporal cortex (FDR). Moreover, Hopp et al. (2023, p.20) find that "political ideology did not significantly moderate neural responses to binding versus individualizing foundations, we did find that activation differences were apparent in regions of the salience and conflict detection network, particularly the ACC and the anterior insular cortex as well as in the left lateral prefrontal cortex (LPFC)." Hopp et al. (2023) acknowledge that this finding needs to be replicated in larger samples - the sample of Hopp et al. (2023) consists of 64 participants (students of the University of California Santa Barbara and people living in the neighborhood). At the same time, Hopp et al. (2023) point out that their findings align, theoretically, with the differences between liberals and conservatives in the ACC that Amodio, Jost, Master, and Yee (2007) documented.

As always, it remains to be seen how robust these preliminary findings with fMRI are. In particular, some of the studies rely exclusively on liberals or people with very strong beliefs, thereby inviting questions regarding generalizability. In addition, neuroscience is moving beyond finding correlations between specific ROIs and outcome variables. In so-called lesion studies participants with brain areas that are absent or damaged are studied to analyze which areas are necessary to perform certain tasks. With this method one can establish the causal effect of certain parts of the brain on reasoning in the brain (Coronel et al., 2012; Nam, Jost, Meager, & Van Bavel, 2021). While lesion studies are by no means "easy" to conduct, we do think that this type of study design can increase our understanding of political information processing.

Second, neuroscience is moving towards identifying broader networks of brain regions that are associated with particular kinds of processing. Hasson, Nir, Levy, Fuhrmann, and Malach (2004) demonstrated that neural activity in the visual cortex is highly correlated across subjects watching the same movie, revealing that individual brains show a surprising tendency to "tick collectively" during natural vision. An example of

an application of this network approach is an analysis of the neural correlates of news sharing identifying regions involved in self-related processing, social processing and valuation as predictors of news sharing (Doré, Scholz, Baek, & Falk, 2020; Scholz et al., 2017). Moreover, van Baar, Halpern, and Feldman Hall (2021) found ideologically aligned brain responses during politically polarized perception.

A different example again relates to ideology. Yang et al. (2022) developed a functional connectivity analysis of 174 liberal and conservative brains, using self-reports and behavioural paradigms to measure performance in various cognitive domains (i.e. retrieval, emotion recognition, empathy, and monetary reward). Their analyses suggest that liberals and conservatives differ in functional connectivity between brain regions during these tasks. The amygdala, inferior frontal gyrus, and hippocampus are most strongly associated with political affiliation. We propose that the analysis of congruent versus incongruent political stimuli could benefit from this network approach.

We are not the first to acknowledge that fMRI research is not easy. It requires access to (very) expensive MRI facilities. The costs per participant are high. And, most of all, there is a steep learning curve for researchers to learn how to design, conduct, analyse and report research that relies upon fMRI data. At the same time, we think that the here discussed research illustrates that fMRI offers insights that other, "relatively" more accessible techniques, cannot provide. Interdisciplinary collaborations within and across universities are one possibility for political psychologists to engage in fMRI research. But embedding rigorous training in neuroscience theories and methods are most likely needed to equip the next generation of researchers with the necessary skills to engage in political neuroscience research. We know this is happening already and welcome such developments.

A look ahead: open theoretical puzzles and methodological improvements

The sections in this chapter, up to this point, do not do justice to the fast amount of research that has been done using skin conductance, fEMG, eye-tracking, heart, EEG and fMRI in neighboring disciplines. It goes without saying that political psychology as a field, has only been scratching the surface when it comes to employing these theories and methods into their toolbox. As we outlined in the discussion, the attempts to "introduce" the measures is not new, nor recent (see for instance, Jost et al., 2014;

Lieberman et al., 2003; McDermott, 2009; Wahlke & Lodge, 1972; Zmigrod & Tsakiris, 2021). At the same time, we have discussed a lot of relatively recent research that relies upon these measures. So perhaps more "innovation" is happening right now in different research labs around the world than what makes it into the journals. We hope that we are indeed correct in observing a move towards more unconscious and preconscious theories and methods in political psychology. If only because politics is too important to leave to the psychologists and neuroscientists. It requires the attention from political psychologists. In the next section, we outline a series of theoretical questions and methodological improvements that hopefully inspire the next generation of research in political psychology.

Comparing outputs of different unconscious measures to understand political information processing

There is relatively little interaction between the research using different psychophysiological and neural measures. There is even less connection with the literature using self-reports. Yet ultimately, we believe, any theory of information processing will benefit from connecting evidence from these different measures. The goal is not necessarily to replicate findings using different measures. Rather each measure may characterize a different process, and jointly we can describe a theory of information processing from the first millisecond of a stimulus to the point of expressing an opinion about the stimulus. We will attempt to do so here by combining our discussions per measure on how people process incongruent versus congruent information (arguments, candidates) in this chapter.

We started with skin conductance. Based on work about the negativity bias (Soroka et al., 2019) we would still hypothesize that processing incongruent information is more arousing than congruent information. At this stage there is very little evidence for this. At best there seems to be inter-individual variation in arousal to political stimuli. We draw the same conclusion regarding research using heart rate variability measures. Then, regarding fEMG, incongruent arguments or politicians (i.e. outparty politicians) produce stronger fEMG activity than congruent arguments or politicians. Depending on the stimuli, corrugator, zygomaticus and labii responses have been reported. Regarding eyetracking it is not yet clear whether we focus on congruent or incongruent political

information, and it may be the case that there are important participant characteristics that moderate these relationships. Comparing different measures used in different research designs is difficult. It is evident we need more similar designs with different psychophysiological measures to more specifically test the tentative hypotheses we set out here.

Moving to neural measures, a few studies using EEG show that incongruent political information produce more specific ERPs and neural oscillations. FMRI work shows that incongruent information produce more left insula and ACC activation. Both approaches stress the early activation of affect in processing political information.

In all it is evident that incongruent information involves both more affective and more cognitive processes. Yet, we believe ultimately the contribution of psychophysiological and neural measures can be to characterize political information processing more. At this stage, the identified affective part of information processing could still reflect affect-as-pressure to update priors, or affect as an in-group defence mechanism to stick to priors.

Alignment between self-reports and unconscious measures

A big, and largely unresolved question for political psychologists is the relationship between the unconscious measures and more explicit self-reported measures.

Political psychologists started with the assumption that there is a direct and unmediated decision-making process that starts with automatic reactions, for example, physiological responses, and ends with behavioral responses (for example, political ideology) (e.g., Aarøe et al., 2017; Friesen, Smith, & Hibbing, 2017; Oxley et al., 2008). Yet, there is research that calls into question the notion of a direct link. At best, physiological and behavioral responses are loosely correlated with each other (Evers et al., 2014; LeDoux & Pine, 2016). The types of distal threatening (negative) stimuli that Oxley et al. (2008), for instance, employed, such as images of spiders, guns and snake, are at best ambiguous indicators of psychological traits. In response to threats, people can regulate the link between automatic responses and cognitive evaluations. Moreover, people can also consider and dismiss threats without engaging in cognitive reflection. Considering these insights, it is not surprising that direct replications failed to find a direct association between physiological indicator of threat sensitivity (as measured

with skin conductance) and self-reported attitudes (Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Fournier et al., 2020; Osmundsen et al., 2022).

Also using fEMG, we found very weak correlations (r<.1) between corrugator responses to incongruent political messages and self-reported negative emotions such as anger and anxiety (Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Bakker et al., 2021). We do find strong correlations between mean fEMG activity and human coding of the valence of images (Schumacher, Homan, et al., 2023) and between self-reported valence and fEMG activity in response to negative vs. positive news (Dubel et al., 2024). We are not alone. Ensari et al. (2004) demonstrate that participants favorably rate politically sensitive out-groups, while their fEMG activity suggests negative affect. In another study, Stewart et al. (2003) examined prejudice towards stigmatized groups, and found no correlation between fEMG activity in response to watching interactions of gay men and self-reported attitudes towards gay men.

In response to these failed replications and the weak correlations between self-reports and physiological indicators of unconscious processes, some have questioned the validity of physiological measures and, in particular, skin conductance. One critique towards skin conductance is based upon the fact that physiological responses to threatening images did not - or very weakly - correlate with self-reported affective responses to the same images (Osmundsen et al., 2022). Yet, we think this lack of alignment is actually not evidence for the lack of validity of one measure over the other. Physiological measures are tapping into other aspects of affect than the self-reports do (Keltner & Gross, 1999). It is therefore not a question of which is "better". These measures are simply tapping into different aspects of affect. So to acquire this rich picture of affect, we also suggest collecting self-reports of emotions because unconscious measurement does not invalidate them.

Moreover, we think that the idea that unconscious measures should "predict" (or correlate) with more explicit attitudes and behavior has put political psychologist on the wrong track. The idea that unconscious processes should predict explicit behaviour, to us, resembles the ongoing discussion between implicitly attitudes - for instance, measured with the Implicit Attitudes Test - and actual attitudes and behaviours (Gawronski, 2019). Implicit attitudes are not considered strong predictors of explicit attitudes and behaviours. So should the unconscious responses we discussed in this

chapter predict explicit political attitudes and behaviours? Perhaps this is not what one should theoretically expect. Inspired by dual-system theories, (Evers et al., 2014, p. 44) explain that "automatic responses are relatively unconscious, fast, and efficient, while reflective responses are relatively conscious, deliberate, and effortful". The reflective system operates slowly and requires effort, while the automatic system is fast and "requires little or no cognitive effort and has a low threshold for processing incoming information" (Evers et al., 2014, p. 44). In their study, Evers et al. (2014) showed that convergence between responses within each of the two systems is larger compared with the coherence in responses between the two systems. Therefore, there is not much reason to expect strong associations between physiological and experiential responses. Turning to questions of interest to political psychologists, self-reported threat sensitivity – as captured with self-reported affect in responses to negative images (Osmundsen et al., 2022), or self-reported personality traits (Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Johnston & Madson, 2022) – correlate stronger with ideology then physiological (Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Osmundsen et al., 2022) or behavioural indicators of threat sensitivity (Johnston & Madson, 2022). In recent work, we are exploring whether physiological responses might correlate stronger with other implicit measures - like implicit ideology assessed with an IAT (Arceneaux et al., 2023). In two laboratory studies in the United States and the Netherlands we do not find evidence that physiological threat sensitivity correlates stronger with implicit measures of ideology (Arceneaux et al., 2023).

Another possibility, that we are currently exploring (Arceneaux et al., 2023), is that under some specific situations, or for some people, implicit (unconscious) and explicit (self-reported) responses align. Let us illustrate this with an example about spiders: people with arachnophobia might consciously be aware that they are afraid of spiders and their unconscious bodily responses (for instance, skin conductance) also show this. Individual differences could perhaps explain this in the domain of politics. Perhaps people who have a better capacity to regulate their emotions interpret their bodily signals better, and might align their unconscious and conscious processes. Aside from individual differences, it could also be that certain situations (an intense political discussion at a birthday party) might align unconscious and conscious responses. At this point, this remains speculation. But we are starting to explore this option

(Arceneaux et al., 2023). Ultimately, we might get a better understanding of the interplay between conscious and unconscious responses. Moreover, perhaps aligned responses have stronger effects on certain political (or non-politics) outcomes.

In fact, we think that one of the big puzzles in the field is to outline when and under which conditions physiological and self-reports align or not. This is echoed by prominent contributions outside of political psychology (e.g., Barrett, Adolphs, Marsella, Martinez, & Pollak, 2019; LeDoux & Pine, 2016). Therefore, we call upon political psychologists to actively engage in what is arguably, one of the biggest puzzles is that researchers of unconscious and more explicit processes face (LeDoux & Pine, 2016). This will not lead to easy answers such as x is a better measure than y is. But ultimately, it will increase our understanding of the processing of political information.

Towards open and transparent research

In various sections of this chapter we have indicated that the design, analysis and reporting of studies using unconscious measures is difficult. Moreover, reflecting on what we have written up to this point, we have mostly discussed confirmatory results: in other words studies that find an effect of a political stimulus on an unconscious measure (or vice versa). Yet, for the literature in political psychology to really make an impact on the literature and society at large, we hope to see a a wider adoption of open and transparent research practices in this literature.

First of all, we need more replications of seminal studies in this emerging field. Science will make progress if we learn more about the replicability and/or generalizability of our foundational studies (McEwan, Carpenter, & Westerman, 2018). This might mean that some of the original findings, including some of our own, might not replicate. But, we should go where the science takes us. As the political scientist Gary King (1995) already pointed out, it would be a good starting point for anybody who is moving into this field (e.g., a graduate student), to replicate and extend the "semininal" study that one wants to build upon. This will immediately create a more robust body of evidence. For instance, we recently successfully replicated and extended the work on the negativity bias (see, Dubel et al., 2024).

Second, we need more preregistered studies. All methods and measures we discussed in this chapter result in complex data that come with a lot of researchers degrees of

freedom (Gelman & Loken, 2013). In fact, working with these measures we started to appreciate the straightforward likert-type survey responses, that for instance measure discrete self-reported emotions (e.g., Rhodes-Purdy, Navarre, & Utych, 2021), more and more. In comparison to survey questions, skin conductance, fEMG, EEG, fMRI data require the researcher to make a lot of decisions throughout the analysis pipeline. As Nosek et al. (2022) explain, preregistration of the analysis pipeline increases the credibility of this literature. Especially for the preprocessing steps and analysis pipeline, we hope to see more discussion about shared standards to analyse unconscious measures, which is happening for instance for fEMG (Schumacher, Homan, et al., 2023), EEG (Baker et al., 2021), and fMRI (Esteban et al., 2019). We hope to see more studies that preregister their hypotheses, design, sample size, and analysis strategy.

Third, an important step in preregistering research is the justification of the sample size. Inspired by Lakens' (2022) call, we urge researchers to justify the effect size of interest. This is inspired by Lakens (2022) who wonders whether it is actually the right question to ask whether studies are sufficiently powered to detect *any* effect? Or to put it differently: "is any effect in the predicted direction ... actually support for the alternative hypothesis" (Lakens, 2022). Political psychologists could follow Primbs et al. (2022) and adopt "a relative framework to effect-size interpretation" where they compare the effects in in their study (or body of studies) to (a) other meta-analytic evidence that employed their specific measure (e.g., skin conductance, fEMG, heart rate, etc) in another field, or (b) other well-known correlates of their outcome variable of interest (e.g., ideology). Something Arceneaux, Bakker, Fasching, and Lelkes (2024) recently tried for the association between personality and political attitudes and showed that the association should be considered small.

The approaches discussed in the previous paragraph allow researchers to interpret the effect sizes in their specific literature to other effect sizes but they "do not quantify which effect sizes are meaningful in specific research lines and, furthermore, there will never be a single answer to the question of which effect size should be considered meaningful" (Anvari & Lakens, 2021). Instead, researchers need to "examine whether observed effects are too small to be theoretically or practically interesting" Lakens (2022). A costs-benefit analysis is one way to do this. In health-research, for instance, researchers calculate whether the costs of an intervention justify the benefits (Anvari &

Lakens, 2021). This might not always be easy for political psychologists. Yet, the goal should be "to determine the smallest increase in a relevant outcome measure that is subjectively deemed to be large enough to matter." Therefore, we join the call made by Arceneaux et al. (2024) and hope to see more studies that define the Smallest Effect Size of Interest (Lakens, 2022) and sufficiently power their studies to reliably detect this effect size, and evaluate their empirical evidence against this bench-mark.

Fourth, we need to make sure that this literature is not suffering from publication bias. Publication bias can happen in, at least, two ways: a) only reporting the studies that worked, and/or b) selectively reporting the hypotheses and/or measures that confirm hypotheses. Selective reporting of findings that "work" significantly bias our understanding of unconscious processes in political psychology. Instead we hope scholars "tell it like it is". One publication format that avoids this form of publication bias is the Registered Reports format. Briefly, this is a two-stage review process in which scholars first submit their theory, design and planned analyses and only upon the acceptance of their "stage 1" manuscript, they collect the data. The first meta-scientific evidence on the adoption of registered reports in psychology suggests that will see more null findings (for preliminary evidence Scheel, Schijen, & Lakens, 2021) but we also get a better understanding of what does and does not work. The journal Politics and the Life Sciences has adopted this publication format. And we think it is also a positive development that Political Psychology, perhaps the flagship journal of the field, is now piloting this publication format in their special issue "Crowd Sourcing the Next Generation of Ideas in Political Psychology", see also: `https://www.next-gen-ideas.com/`). We hope that more scholars adopt the registered report format when employing the theories and methods that we discussed in this paper.

These four suggestions to make the field more open and transparent will ultimately increase the replicability and credibility of political psychology research that relies upon unconscious measures.

Getting more N, getting more variation

Many of the papers we have discussed here have a low number of participants and typically low ideological variation. For example, the Kanai et al. (2011) study has not a single participant at point 5 of their 1-5 scale measuring conservatism. Often it is

difficult to find right-wing participants on overwhelmingly left-wing university campuses. Recruiting participants off-campus is often expensive, limited due to travel time, and we expect some people are unwilling to visit left-wing campuses. Ultimately, findings would become more robust if we can increase the N, and increase the ideological variation of studies using psychophysiological or neural measures. But how to do this? In this section we discuss a few of these options.

One option is to set up a *lab-in-the-field*. There are several examples of researchers going to malls, museums, festivals to recruit participants, and using an on-site location or a truck to conduct the study. In our own experience targeting different sorts of festivals or museums with a broad appeal, diversified our sample particularly by attracting more right-wing participants (Bakker et al., 2021; Schumacher, Homan, et al., 2023). It remained difficult to attract populist voters (Schumacher et al., 2022). At the same time labs-in-the-field also bring challenges: you can only field very short experiments, because people are unlikely to commit much time. People may have been drinking or have used drugs - although our results seem unaffected when controlling for alcohol levels (measured using a breathalizer) and drugs intake (with a self-reported measure) (Bakker et al., 2021). Circumstances like temperature are more difficult to control outside of the lab. In our experience there are practical solutions to many of these issues, and sometimes some issues - liking drinking alcohol - may in fact be closer to how we process political information in the wild. Ultimately, less perfect circumstances may create more noise in your data, but having a more diverse ideological sample is in many cases worth it especially if you expect or try to find evidence for ideological asymmetries.

A different option is to measure psychophysiological measures *online* using emerging technologies. Online you can collect eyetracking data, heart rate and heart rate variability and recognize emotions using facial muscle movement (Boussalis, Coan, Holman, & Müller, 2021; Fridkin, Gershon, Courey, & LaPlant, 2021; Masch, Gassner, & Rosar, 2021). Regarding the latter, several examples of so-called emotion recognition technology have emerged: (1) FaceReader (Noldus Information Technology), a licensed algorithm developed for academia, (2) Microsoft Azure Cognitive Services' Face API, a commercial algorithm with limited free access, and (3) the open-source algorithm OpenFace. Most of these emotion recognition algorithms are trained on the Facial

Action Coding System (FACS), a detailed rule book for human coders to detect emotional expressions in the face (Ekman & Friesen, 2003). The FACS is based on Action Units, which represent each individual facial muscle that, in combination, are assumed to represent certain emotional expressions. FACS coding requires intensive training, is expensive and time-consuming, making these automated emotion recognition algorithms more appealing. In addition, these algorithms have a very high accuracy when compared to FACS coding (Perusquía-Hernández, Ayabe-Kanamura, Suzuki, & Kumano, 2019). Comparisons between emotion recognition algorithms and fEMG are however rare, due to the fact that in most setups the action units are obscured by the fEMG electrodes. Using new distal fEMG technology Perusquía-Hernández et al. (2019) compared the two measures. Their conclusion is that fEMG outperforms emotion recognition algorithms in covert muscle activation. However, the algorithms outperform fEMG in overt muscle activation. On this basis we suggest that for research questions about how individuals process information fEMG is preferable over emotion recognition algorithms because participant responses are likely to be more covert. Yet, when analyzing elite interactions we suggest researchers to use emotion recognition algorithms instead, because such interactions include more overt affect signalling and of course because it is feasible. Moreover, most algorithms are trained to a large extent with data from actors, so it should work better in analyzing public performances than in affective responses in the lab.

So far our suggestions to increase N and increase ideological diversity have concentrated on designs similar to the standard lab experimental design. Yet these designs typically lack ecological validity (Bronfenbrenner, 1979; Nastase, Goldstein, & Hasson, 2020). It is not natural to read, watch or listen to news (or other media content) while sitting in an artificial (almost hospital-like) laboratory environment while connected to a set of wires to your body (in the case of physiology) or even laying down in a large and noisy magnet (in case of fMRI). To kill two birds (low N, high artificiality) with the same stone, we suggest looking into *wearables*. Wearables can measure skin conductance, heart rate and can be easily connected to questionnaires (Bolinski et al., 2021; Jahn, Meshi, Bente, & Schmälzle, 2022; Konvalinka et al., 2011). To a more limited extent there is also a mobile version of EEG (Jahn et al., 2022). Using wearable devices, it becomes possible to study people at home, at school, among

their friends, or at any other place where people process political content. There is a lot of unexplored ground for political psychology to study unconscious responses to politics outside the classical laboratory setting, during for example political rallies, election debates, protests, or in other more ecologically valid environments in which people are exposed to politics. Such ecologically more valid studies will make important contributions to our understanding of the unconscious responses to politics.

Finally, we hope that political psychology continues to move towards a "team science" approach. Instead of each of us continuing to collect data in our individual labs, we hope that more multi-laboratory collaborations will start. This is not easy. It requires standardized equipment across laboratories. It requires a willingness to devote costly laboratory time to a collaboration with others. Most of all, it requires intensive collaborations. We believe that the costs outweigh the benefits: sample sizes become bigger, cross-laboratory variation can be utilized, and designs of studies will become better by collaborating across laboratories. Cross-laboratory collaborations have proven to be fruitful (for exmample, Bakker, Schumacher, Gothreau, & Arceneaux, 2020; Osmundsen et al., 2022). Ultimately, these cross-laboratory collaborations also increase diversity in the contexts that are being studies. This is important, as we would be the first to acknowledge that a lot of the literature discussed in this chapter rely upon samples conducted among narrow subsets (often students) living in Western, Educated, Industrialized, Rich and Democratic countries. The ground-breaking cross-cultural work by Soroka et al. (2019) should be an example and inspiration to all of us.

How to effectively bridge Politics and Psychology?

Throughout this chapter we hinted at obstacles scholars face if they want to use the methods we discussed. These obstacles can be summarized as follows: 1) starting a lab is expensive especially if your university or department does not have infrastructure. 2) If you are new into this field you need training in designing the studies, collecting the data, analysing the data and reporting it. 3) There is discussion, especially in the political science pillar of political psychology, what the validity of some of these measures is as they correlate weakly with the much beloved self-reports. 4) Publishing interdisciplinary research is hard. At conferences and workshops those new into this line of research are often exposed to discouraging stories of those who "tried this before" but

"failed" (getting this work published). That said, we do not think these four reasons are compelling enough to stop us from encouraging others to join us in the pursuit we outlined in this chapter.

Are you still on board? If so, we will now share some of the lessons we learnt (the hard way). We will distinguish between lessons for political psychologists trained in the tradition of (mostly) survey-based research and neuroscientists with more experience in lab-based paradigm-oriented work. A straightforward lesson is that you need to find collaborators. But what disciplinary boundaries do you encounter?

First, research designs differ. Political psychology uses mostly large-n survey-based research with embedded between-subjects experiments. In neuroscience within-subjects designs are much more common. There are several extensively validated paradigms used in neuroscience that may also be beneficial to political psychology.

Second, political psychologists and neuroscientists need to acknowledge that politics is complex. Ideology, for instance, cannot be captured with one self-reported item asking people on a scale from 1 to 10 how left-wing or right-wing they are (Malka, Lelkes, & Soto, 2019). Moreover, what might "work" in a two-party context, may or may not, translate to another political context. At the same time, psychological constructs do not always need to be measured with very extensive batteries (Bakker & Lelkes, 2018).

Third, there is a lot of potential to design studies that build upon existing paradigms in neuroscience. We encourage researchers to develop a politicized version of the original paradigm, and use a design in which both the original and new, politicized version are tested. For example, we exposed participants both to images of non-political individuals with specific facial displays, as well as images of political leaders with specific facial displays Homan, Schumacher, and Bakker (2023). In another study we manipulated the beads task. In this task participants need to guess the proportion of colored marbles in a jar, and they are given the option to continuously draw more marbles from the jar before deciding. After running this task, we exposed participants to a politicized version in which the marbles became votes for different political parties Petropoulus Petalas, Bakker, and Schumacher (2022). Following this setup, researchers can first assess if the basic psychological mechanism is indeed replicated and directly speak to the neuroscientific (psychological) theory. Then, they contrast the findings from the non-political condition to the political condition. This way, they can assess if

and to what extent politics moderates information processing.

Fourth, political psychologists and neuroscientists need to be aware that research and publication cultures differ. Reading across disciplinary boundaries is a good start to get an impression *how* scholars communicate with each other in different fields. Talking about potential outlets at the start of a project is useful. Another aspect of the publication culture is that neuroscientists often work in larger teams where different people play a part in the research process. This, again, is less common in political psychology. The use of the CRediT taxonomy (https://credit.niso.org/) can be helpful here to clearly specify who did what in the project.

Ultimately, we think there is a lot of potential for a "beautiful friendship" (Jost et al., 2014). As we have discussed in this chapter, there is increasing evidence that politics is a unique context for the study of human psychology. To further flesh this out, we need long-term collaborations between researchers from different backgrounds. Perhaps this could even lead to findings (or even theories) that would be of interest to the broader community of researchers interested in human psychology. The interdisciplinary field of political psychology should have that ambition.

Where will political psychology be in 10 years?

This handbook set out to document "innovations in political psychology". We have tried to document the innovations when it comes to unconscious processes. We are optimistic that the theories and methods we describe can increase our understanding of political information processing. This is not a novel argument. Yet, we hope our chapter won't be the next citation in a series of contributions (e.g., Cacioppo & Visser, 2003; Jost et al., 2014; Lieberman et al., 2003; McDermott, 2009) that have documented the "promise" of combing political science with insights from neuroscience. Instead, we hope readers of this chapter take up the challenge to systematically study the unconscious processes that must shape political information processing. The questions that modern societies face, ranging from polarisation to misinformation and from political disaffection to political violence are too important to only study with the insights that self-reported attitudes and behaviours provide us. At the same time, we acknowledge that a single unconscious measure will not solve these complex societal issues. But using the analogy of a flashlight in the dark: limiting ourselves to

self-reports, like political scientists predominantly do (see, Webster & Albertson, 2022), we only get answers to our research questions at the place where we direct our flashlight. If we bring theories and methods of unconscious measures into our toolbox, the beam of the flashlight would be a lot wider and we would be getting more insights. If our goal is to increase our understanding of political information processing, then this is where the field should be heading towards. This will not be easy and it will require each of us to learn new skills, form new collaborations, and accept that we might fail at times. But ultimately, we increase the chances of getting a better grip on the complex behaviour of humans in the domain of politics.

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